

⑥ Register data type

- in verilog a 'register' is a variable that can 'Hold' a value.
 - unlike 'wire' that is continuously driven and ~~can~~ hold any value.
 - does not necessarily mean that it will map to a hardware register during synthesis.
 - combinational circuit specification can also use register type variables.
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• Register data types supported by verilog:

- i. reg : most widely used
- ii. integer : used for loop counting (typical use)
- iii. real : used to store floating-point number
- iv. time : keeps track of simulation time (not used in synthesis)

• "reg" data type:

- Default value of a 'reg' data type is 'x'.
- It can be assigned a value in synchronism with a clock or even otherwise.
- the declaration explicitly specifies the size (default is 1-bit):
reg x, y; // single-bit register variable

reg [15:0] bus; // A 16-bit bus

- Treated as an unsigned number in arithmetic expression.
- must be used when we model actual sequential hardware element like counters, shift registers, etc.

A = B + C ; A - reg type variable

A <= B + C ;

module simple_counter (clk, rst, count);

input clk, rst;

output [31:0] count;

reg [31:0] count;

always @ (posedge clk)

begin

if (rst)

count = 32'b0;

else

count = count + 1;

end

endmodule

32-bit counter with synchronous reset.

• count value increases at the positive edge of the clock.

• If 'rst' is high the counter is reset at the positive edge of the next clock.